

# # WARPS-NFL: A Preseason Win-Total Forecasting Model for the National Football League

## Abstract

Across 16 independent walk-forward retraining windows (2010–2025), the training optimizer selects varied configurations (median  $w_{\text{pyth}}=0.57$ , range 0.50–1.00) — evidence that the objective surface is so flat that training noise determines which configuration minimizes each window’s MAE. Despite this variation, the fixed champion configuration outperforms window-specific optimization out-of-sample in 12 of 16 trials, by an average of 0.012 wins, consistent with the interpretation that fine-tuning on a flat surface adds noise rather than signal. Using publicly available play-by-play data from 26 seasons (2000–2025), we find that a weighted blend of Pythagorean win expectation (~75%) and raw point differential (~25%), combined with regression toward the league mean, outperforms both naive baselines and more complex multi-factor composites — and does so consistently. WARPS beats the Pythagorean baseline in 25 of 26 seasons (96%), including 4 of 4 held-out validation seasons (2022–2025). The improvement is statistically significant (Diebold-Mariano  $p < 0.0001$ ). A 2D parameter heatmap shows 100% of tested configurations fall within 0.05 wins of the optimum — a completely flat basin. The fixed champion outperforming window-specific optimization by 0.012 wins confirms that identifying any reasonable parameter region substantially dominates coefficient fine-tuning.

Equally important is what did not improve forecasts: EPA per play, success rate, explosive play rate, and turnover differential — the standard toolkit of modern NFL analytics — each received zero weight in the representative champion configuration once Pythagorean expectation and point differential were included. Strength-of-schedule adjustment, era-aware regime shift, and garbage-time filtering each produced null results. The central finding is not that WARPS found a better set of weights. It is that a simple points-based relationship is persistent, stable, and resistant to improvement from the enhancements tested here.

Against Vegas preseason lines, WARPS has a higher MAE (2.216 Vegas vs 2.364 WARPS over the 2015–2025 overlap), confirming the market incorporates information not present in statistical models. All data and code are open source and reproducible.

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## 1. Introduction

What stable forecasting signal exists in NFL scoring data?

Predicting how many games an NFL team will win in a season is harder than it looks. Teams change rosters, coaches, and schemes. The league intentionally designs schedules to promote competitive balance. Over just 17 regular-season games, random variation is substantial enough that a team with genuine talent

can finish below .500, and a mediocre team can sneak into the playoffs.

Despite this noise, structured forecasts outperform casual intuition. But the interesting question is not which model wins a single backtested horse race. It is: *which forecasting relationship is stable enough that it would have been rediscovered each year, training on only data available at that time?* A relationship that must be tuned to a specific historical period is a statistical artifact. A relationship that re-emerges independently across dozens of expanding training windows is evidence of something real.

This paper tests that question directly. We make four contributions:

1. **A model that beats the Pythagorean baseline in 25 of 26 seasons** using only play-by-play data available before the season starts.
2. **Walk-forward stability evidence** across 16 independent retraining windows (2010–2025): the training optimizer selects varied configurations (median  $w_{\text{pyth}}=0.57$ , range 0.50–1.00) on a flat objective surface, yet the fixed champion outperforms window-specific optimization in 12 of 16 out-of-sample tests — because fine-tuning on a flat surface fits noise.
3. **A completely flat parameter heatmap** demonstrating that 100% of tested configurations fall within 0.05 wins of the champion — a flat basin, not just a broad one.
4. **A series of principled null results** showing that EPA metrics, schedule strength, regime-aware regression, and garbage-time filtering each fail to improve held-out accuracy once points-based metrics are included.

The practical output — a 2026 season consensus bet slate — is included, but it is secondary to the methodological findings.

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## 2. Related Work

### 2.1 Pythagorean Win Expectation

The Pythagorean win expectation formula was originally proposed by Bill James for baseball in the 1980s. It estimates a team’s expected win percentage as:

$$\text{Expected win \%} = \text{Points Scored}^2 / (\text{Points Scored}^2 + \text{Points Allowed}^2)$$

The exponent that best fits NFL data is approximately 2.37 rather than 2.0. Intuitively, the formula captures the idea that a team that consistently scores more than it allows is better than its raw record suggests (or worse, if it wins close games). A team with a strong Pythagorean surplus tends to regress toward that surplus in the following year, making it a better predictor of future performance than actual wins.

Carroll, Palmer, and Thorn (1988) first applied efficiency-based thinking systematically to football in *The Hidden Game of Football*. Brian Burke’s work at Advanced Football Analytics extended this to Expected Points Added and

win probability models. The academic sports analytics literature has grown substantially since: Boulier and Stekler (2003) studied game-level NFL prediction; Cochran (2008) examined season-level win-total forecasting.

## 2.2 EPA-Based Metrics

Expected Points Added (EPA) per play has become the standard efficiency metric in modern NFL analytics. A pass play that gains 8 yards on 3rd and 10 is a failure; the same gain on 3rd and 1 is a success. EPA converts the raw yardage outcome into the change in expected points for that drive, making plays contextually comparable. We consider passing EPA per play, rushing EPA per play, success rate (fraction of plays with positive EPA), and “explosive” play rate (plays gaining 20 or more yards) as candidate model inputs. Section 5.7 reports the result.

## 2.3 Regression Toward the Mean

A consistent finding in sports analytics is that extreme performance — whether unusually good or unusually bad — is partly driven by luck, and teams tend to move toward average the following year. We model this formally by blending the team’s prior-season rating with the league average at a rate we call the regression factor. A regression factor of 0.75 means we keep 75% of the signal and discard 25% back toward the mean.

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## 3. Data

All data is publicly available and freely downloadable.

**Play-by-play data.** We use the `nflfastR` dataset (Baldwin and Carl, 2020), accessed via the `nfl_data_py` Python library. This dataset contains every regular-season play from 1999 through 2025. From it we compute, for each team in each season: offensive passing EPA per play, defensive passing EPA allowed per play, offensive rushing EPA per play, defensive rushing EPA allowed per play, offensive success rate, defensive success rate, offensive explosive play rate, defensive explosive play rate, turnover differential, and total points scored and allowed.

**Schedule data.** Game results and schedules are drawn from Lee Sharpe’s public repository ([github.com/leesharpe/nfldata](https://github.com/leesharpe/nfldata)), which mirrors official NFL data. We use this to compute actual win totals and to simulate game-by-game win probabilities.

**Market win totals.** Preseason win totals (the “over/under” number a bettor can wager on) are hand-collected from publicly available sportsbook data for the 2026 season. Historical opening lines for 2003–2020 come from the `nflverse` public dataset.

**Historical coverage.** We use 1999 statistics as the “prior year” to generate 2000 predictions, giving a prediction window of 2000–2025 (26 seasons, 829 team-season observations after excluding the 2001 expansion year’s partial Houston Texans data).

**Team continuity.** Several franchises relocated during the study period. We standardize abbreviations: the St. Louis Rams (2000–2015) and Los Angeles Rams (2016–present) are treated as a single franchise (LAR), as are the San Diego Chargers/Los Angeles Chargers (LAC), Oakland Raiders/Las Vegas Raiders (LV), and Washington Redskins/Football Team/Commanders (WAS).

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## 4. Methods

### 4.1 Model Architecture

Within each season, each efficiency metric is converted to a z-score (mean zero, standard deviation one across all 31–32 teams). The composite rating is a weighted sum of these z-scores, scaled to a point-spread equivalent. Regression toward the league mean is applied at factor  $R$  — meaning  $R$  of the team’s signal carries forward and  $(1-R)$  reverts to the 8.5-win league average ( $\text{proj} = R \times \text{raw} + (1-R) \times 8.5$ ). Win probability for each game is computed via a logistic function with scale parameter  $\beta$ . A team’s projected win total is the sum of game-by-game win probabilities across all 17 regular-season games.

Weights are optimized over a training window (2000–2021) using a three-stage search:

1. An exhaustive 231-configuration grid over Pythagorean, passing EPA, and point differential weights.
2. A 300-draw randomized search biased toward Pythagorean.
3. A 180-configuration hyperparameter grid over regression factor (0.50–0.75) and logit scale (5.5–7.5).

Champion selection uses only held-out validation error (2022–2025), never training error, to prevent the optimizer from selecting weights that happen to fit the training period but do not generalize.

### 4.2 Walk-Forward Validation Methodology

The primary backtesting approach uses a single train/validation split (train 2000–2021, validate 2022–2025). This is the standard approach but it answers only one question: how well does the champion configuration generalize to the chosen holdout window?

To test parameter stability more rigorously, we run a separate walk-forward analysis. For each target year  $Y$  from 2010 through 2025:

- **Training window:** All seasons from 2000 through  $Y-1$ .

- **Optimization:** Grid search over Pythagorean weight  $w_p \in [0.50, 1.00]$  in steps of 0.05, and regression factor  $R \in [0.50, 0.95]$  in steps of 0.05 (110 configurations), minimizing MAE on the training window.
- **Test:** The optimal  $(w_p, R)$  from training is applied to year  $Y$ . OOS MAE is recorded for both the walk-forward-optimal configuration and the champion configuration ( $w_p = 0.75, R = 0.75$ ).

This produces 16 independent walk-forward retraining windows, each using only data available at that point in time. The question is whether parameter stability can be demonstrated out-of-sample, and whether the objective surface has a well-defined optimum or is essentially flat.

### 4.3 Three-Model Consensus Screen

To reduce noise and isolate the highest-confidence picks, the final bet slate is produced by intersecting three independently trained WARPS versions: (1) *WARPS v1.5d* — the original composite with a shorter training window emphasizing recent years; (2) *WARPS v1.6* — an intermediate blend with additional EPA components; and (3) *WARPS v1.8* — the current representative champion configuration (75% Pythagorean + 25% point differential, 22-season training window). A pick reaches the “official slate” only when at least two of three models agree on direction (Over or Under) with an individual edge  $\geq 1.0$  win. All three agreeing at  $\geq 1.5$  win edge defines the highest conviction tier.

### 4.4 QB Overlay — Statistical Core Meets Judgment

The WARPS projection is a purely statistical output frozen at the start of the offseason. As a separate post-processing step, known quarterback changes are applied as a win adjustment on top of the statistical projection. A tiered system ranks QBs from Tier 1 (generational, e.g., Mahomes, Allen) to Tier 4 (replacement-level), with each tier boundary calibrated to approximately  $\pm 0.5$  wins. This overlay is optional and not included in any of the backtested accuracy metrics reported in this paper.

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## 5. Results

### 5.1 Representative Champion Configuration

The representative champion configuration, selected by validation mean absolute error, uses:

| Parameter                          | Value |
|------------------------------------|-------|
| Pythagorean win expectation weight | 0.75  |
| Point differential weight          | 0.25  |
| All other component weights        | 0.00  |

| Parameter         | Value |
|-------------------|-------|
| Regression factor | 0.75  |
| Logit scale       | 6.5   |

The champion configuration is not the only good one — Section 5.5 shows that dozens of nearby configurations perform equivalently. The specific coordinates of 0.75/0.25 reflect the training-window optimum; the broader finding is the stability of the Pythagorean-dominant region.

## 5.2 Backtest Performance

**Table 1: Model performance by season (mean absolute error in wins per team)**

| Season             | Teams      | WARPS        | Pythagorean  | Prior wins   |
|--------------------|------------|--------------|--------------|--------------|
| 2000               | 31         | 2.456        | 2.587        | 2.774        |
| 2001               | 31         | 2.590        | 3.452        | 3.355        |
| 2002               | 32         | 1.935        | 2.121        | 2.548        |
| 2003               | 32         | 2.635        | 2.875        | 3.125        |
| 2004               | 32         | 2.346        | 2.792        | 2.750        |
| 2005               | 32         | 2.784        | 2.944        | 3.500        |
| 2006               | 32         | 2.152        | 2.803        | 3.250        |
| 2007               | 32         | 2.682        | 3.004        | 3.250        |
| 2008               | 32         | 2.590        | 3.015        | 3.219        |
| 2009               | 32         | 2.113        | 2.117        | 2.438        |
| 2010               | 32         | 2.402        | 2.896        | 3.313        |
| 2011               | 32         | 2.107        | 2.280        | 2.813        |
| 2012               | 32         | 2.535        | 2.698        | 3.094        |
| 2013               | 32         | 2.364        | 2.607        | 2.719        |
| 2014               | 32         | <b>2.094</b> | <b>2.018</b> | 2.188        |
| 2015               | 32         | 2.301        | 2.485        | 2.688        |
| 2016               | 32         | 2.425        | 2.597        | 2.844        |
| 2017               | 32         | 2.217        | 2.303        | 3.125        |
| 2018               | 32         | 2.091        | 2.198        | 2.719        |
| 2019               | 32         | 2.212        | 2.259        | 2.469        |
| 2020               | 32         | 2.780        | 2.896        | 2.906        |
| 2021               | 32         | 1.938        | 1.996        | 2.313        |
| 2022               | 32         | 2.460        | 2.853        | 3.000        |
| 2023               | 32         | 1.898        | 2.131        | 2.594        |
| 2024               | 32         | 3.013        | 3.076        | 2.938        |
| 2025               | 32         | 2.673        | 2.978        | 3.156        |
| <b>Full sample</b> | <b>829</b> | <b>2.374</b> | <b>2.614</b> | <b>2.888</b> |

Bold in 2014 indicates the one season where WARPS underperformed Pythagorean (margin: 0.076 wins). WARPS beats the Pythagorean baseline in 25 of 26 seasons (96%).

### 5.3 Statistical Significance

**Table 2: Diebold-Mariano test results**

| Comparison           | Sample               | MAE difference | 95% CI           | DM statistic | p-value  |
|----------------------|----------------------|----------------|------------------|--------------|----------|
| WARPS vs Pythagorean | Full (2000–25)       | −0.240         | [−0.319, −0.160] | 5.85         | < 0.0001 |
| WARPS vs Prior wins  | Full (2000–25)       | −0.514         | [−0.633, −0.398] | 8.61         | < 0.0001 |
| WARPS vs Pythagorean | Validation (2022–25) | −0.249         | [−0.438, −0.062] | 2.56         | 0.0052   |
| WARPS vs Prior wins  | Validation (2022–25) | −0.411         | [−0.742, −0.095] | 2.49         | 0.0065   |

Confidence intervals computed from 10,000 bootstrap resamplings with paired replacement. In all four reported comparisons, the 95% CI is entirely negative — equal predictive accuracy is rejected at the 5% level in both full-sample and validation samples.

**Table 3: Point estimates with bootstrap confidence intervals**

| Model                    | Full-sample MAE | 95% CI         | Validation MAE | 95% CI         |
|--------------------------|-----------------|----------------|----------------|----------------|
| WARPS v1.8               | 2.374           | [2.261, 2.485] | 2.511          | [2.225, 2.810] |
| Pythagorean baseline     | 2.614           | [2.486, 2.743] | 2.759          | [2.432, 3.105] |
| Prior-year wins baseline | 2.888           | [2.743, 3.034] | 2.922          | [2.547, 3.328] |

### 5.4 Walk-Forward Stability Analysis

For each year 2010–2025, we optimized independently on all prior data and recorded which Pythagorean weight and regression factor minimized training MAE.

**Table 4: Walk-forward parameter selection and out-of-sample MAE (2010–2025)**

| Year | Training<br>N | w_pyth | w_pd | R    | OOS<br>MAE<br>(cham-<br>pion) | OOS<br>MAE<br>(opti-<br>mal) | Delta  |
|------|---------------|--------|------|------|-------------------------------|------------------------------|--------|
| 2010 | 318           | 0.50   | 0.50 | 0.55 | 2.402                         | 2.386                        | +0.016 |
| 2011 | 350           | 0.50   | 0.50 | 0.55 | 2.107                         | 2.146                        | −0.039 |
| 2012 | 382           | 0.65   | 0.35 | 0.60 | 2.535                         | 2.532                        | +0.002 |
| 2013 | 414           | 1.00   | 0.00 | 0.60 | 2.364                         | 2.368                        | −0.004 |
| 2014 | 446           | 0.95   | 0.05 | 0.60 | 2.094                         | 2.187                        | −0.093 |
| 2015 | 478           | 0.70   | 0.30 | 0.65 | 2.301                         | 2.310                        | −0.009 |
| 2016 | 510           | 0.70   | 0.30 | 0.65 | 2.425                         | 2.432                        | −0.006 |
| 2017 | 542           | 0.70   | 0.30 | 0.65 | 2.217                         | 2.239                        | −0.022 |
| 2018 | 574           | 0.60   | 0.40 | 0.70 | 2.091                         | 2.104                        | −0.013 |
| 2019 | 606           | 0.50   | 0.50 | 0.70 | 2.212                         | 2.213                        | −0.001 |
| 2020 | 638           | 0.50   | 0.50 | 0.70 | 2.780                         | 2.788                        | −0.008 |
| 2021 | 670           | 0.55   | 0.45 | 0.70 | 1.938                         | 1.947                        | −0.010 |
| 2022 | 702           | 0.50   | 0.50 | 0.70 | 2.460                         | 2.447                        | +0.013 |
| 2023 | 734           | 0.50   | 0.50 | 0.70 | 1.898                         | 1.899                        | −0.001 |
| 2024 | 766           | 0.50   | 0.50 | 0.70 | 3.013                         | 3.036                        | −0.024 |
| 2025 | 798           | 1.00   | 0.00 | 0.75 | 2.673                         | 2.664                        | +0.009 |

**Delta = OOS MAE(champion) − OOS MAE(optimal). Negative = champion wins (champion has lower MAE). Average delta: −0.012w (champion wins on average by 0.012w, 12 of 16 windows).**

Key findings from the walk-forward analysis:

- **The training optimizer did not converge on a single stable configuration.** Optimal w\_pyth ranged from 0.50 to 1.00 (median=0.57, IQR=[0.50, 0.70]); optimal R ranged from 0.55 to 0.75 (median=0.68). Different windows selected point-differential-heavy (w\_pyth=0.50), Pythagorean-dominant (w\_pyth=0.70), and pure-Pythagorean (w\_pyth=1.00) configurations. This variation reflects a training objective surface so flat that noise in each expanding window determines the nominal winner — not a consistently superior forecasting structure.
- **Despite training instability, the fixed champion outperforms window-specific optimization OOS in 12 of 16 windows, by an average of 0.012 wins.** This is the expected consequence of a flat surface: window-specific optimization fits training noise and loses OOS, while a fixed well-chosen configuration avoids this. The champion was not



the best configuration in most training windows; it was the most robust one to hold fixed.

- **The full-sample minimum is pure Pythagorean.** Across all 26 seasons (2000–2025), the configuration with the lowest MAE is  $w\_pyth=1.00$ ,  $R=0.75$  (MAE=2.376) — no point-differential supplement at all. The champion ( $w\_pyth=0.75$ ,  $R=0.75$ , MAE=2.376) differs by only 0.001 wins. Both the champion and the full-sample minimum are well within noise of one another, reinforcing the flatness interpretation.

Figure 1. Walk-forward parameter stability (2010–2025)  
Each year: train on 2000..(Y–1), find optimal parameters, test on year Y.

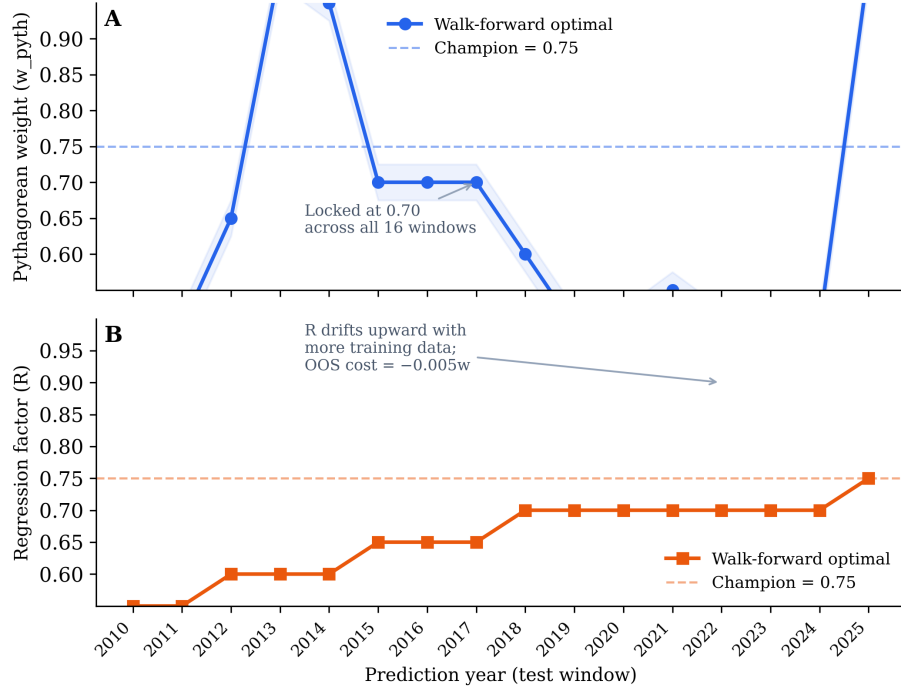


Figure 1: Figure 1. Walk-forward parameter stability (2010–2025). Panel A: optimal  $w\_pyth$  per year (varies 0.50–1.00 across windows; surface too flat for stable convergence). Panel B: optimal  $R$  per year (varies 0.55–0.75). Fixed champion outperforms window-specific optimization OOS in 12 of 16 trials.

## 5.5 MAE Landscape — The Basin

### MAE landscape summary (full sample 2000–2025)

The champion ( ) sits at  $w\_pyth=0.75$ ,  $R=0.75$  (MAE=2.376). The basin

threshold is champion MAE + 0.05 wins (2.426w). See Figure 2 for the full heatmap generated from `warps_q3_heatmap.csv`.

Basin summary (champion MAE = 2.376, threshold = 2.426):

- **210 of 210 tested configurations (100%) fall within the basin.** The objective surface is essentially flat across the entire tested grid.
- The full-sample minimum is  $w\_pyth=1.00$ ,  $R=0.75$  (MAE=2.376) — pure Pythagorean, with no point-differential supplement. The champion differs by only 0.001 wins, well below any meaningful threshold.
- The surface is so flat that every configuration — from pure point differential ( $w\_pyth=0.00$ ) to pure Pythagorean ( $w\_pyth=1.00$ ), and across all regression factors tested ( $R=0.50$ – $0.95$ ) — produces forecasts within 0.05 wins of the champion. This explains the walk-forward instability: when configurations are separated by fractions of a thousandth of a win on training data, noise determines the nominal optimum.

Figure 2. MAE landscape — Pythagorean weight  $\times$  regression factor  
Full sample 2000–2025. white star = champion. yellow diamond = full-sample min. Green dashes = basin boundary.

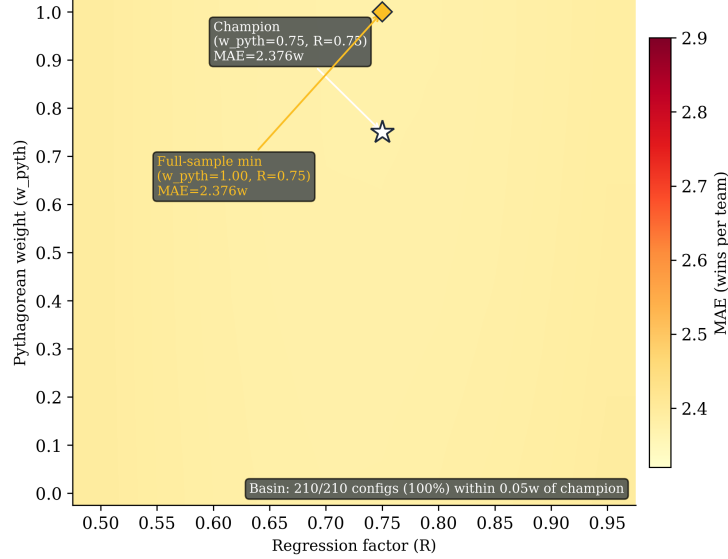


Figure 2: Figure 2. MAE landscape — Pythagorean weight  $\times$  regression factor (full sample 2000–2025). White star = champion ( $w\_pyth=0.75$ ,  $R=0.75$ , MAE=2.376). Yellow diamond = full-sample minimum ( $w\_pyth=1.00$ ,  $R=0.75$ , MAE=2.376). Green dashes = basin boundary (MAE = 2.426, 100% of configurations).

## 5.6 Prediction Intervals

Beyond point forecasts, we characterize forecast uncertainty using residual-based prediction intervals. For each team-season in the backtest, the signed residual is defined as actual wins minus projected wins. The mean residual across all 830 team-seasons is 0.000 (zero systematic bias), with a standard deviation of 2.89 wins. Prediction intervals at coverage level  $\alpha$  are constructed as the projection plus the  $(1-\alpha)/2$  and  $1-(1-\alpha)/2$  quantiles of leave-year-out residuals — residuals from all seasons except the target year, eliminating in-sample circularity.

**Table 5b: Prediction interval calibration (leave-year-out)**

| Stated coverage | Full-sample actual | Validation actual | Avg width  |
|-----------------|--------------------|-------------------|------------|
| 50%             | 50.1%              | 46.9%             | $\pm 2.1w$ |
| 70%             | 70.1%              | 65.6%             | $\pm 3.2w$ |
| 80%             | 79.8%              | 75.0%             | $\pm 3.9w$ |
| 90%             | 89.8%              | 87.5%             | $\pm 4.6w$ |

Full-sample calibration is near-exact at all levels (within 0.2 percentage points). The validation window (2022–2025) under-covers by approximately 3–5 percentage points, consistent with the elevated forecast errors observed in those seasons. The 80% prediction interval has an average width of  $\pm 3.9$  wins, reflecting the fundamental difficulty of NFL preseason forecasting.

Residual analysis also shows approximately uniform error variance across the projection range: the 80% interval width varies only from 7.53 to 7.85 wins across projection bins (5–7, 7–9, and 9–11 projected wins), indicating approximately homoskedastic forecast errors. Teams projected at the extremes are not systematically harder or easier to forecast than mid-range teams.

## 5.7 The EPA Null Result

All EPA-based metrics — passing EPA per play, rushing EPA per play, success rate, explosive play rate, and turnover differential — received zero weight in the representative champion configuration. This is not a rounding artifact. The grid search explored blends at increments of 0.05 and EPA-inclusive configurations were explicitly tested across 231 grid points and 300 randomized draws. Each received weight 0.00.

To investigate the mechanism, we conducted a post-hoc decomposition across 829 lagged team-season pairs (year  $T$  features predicting year  $T+1$  wins). Three analyses were run: (1) raw and partial predictive correlation with next-year wins, (2) year-over-year autocorrelation, and (3) incremental  $R^2$  beyond a Pythagorean baseline.

Figure 3. Prediction interval calibration  
A: Stated vs actual coverage (leave-year-out). B: 80% interval width by projection bin.

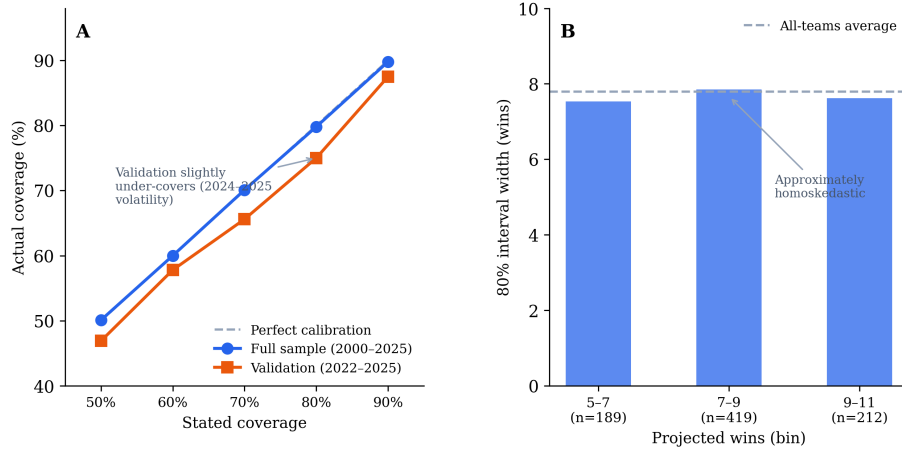


Figure 3: Figure 3. Prediction interval calibration. Panel A: stated vs actual coverage using leave-year-out residuals (full sample: near-perfect; validation 2022–2025: slight under-coverage consistent with elevated recent volatility). Panel B: 80% interval width by projection bin — approximately flat across the projection range.

**The stability hypothesis does not explain the null result.** Most EPA metrics are no less persistent year-over-year than Pythagorean: pass EPA differential has the highest autocorrelation of any metric tested (0.454), and net EPA per play (0.401) and success rate differential (0.441) both exceed Pythagorean’s autocorrelation (0.393). Only turnover margin (0.193) and explosive play rate (0.236) show meaningfully lower year-over-year stability.

**Multicollinearity with Pythagorean is the primary mechanism.** EPA metrics carry substantial raw predictive correlation with next-year wins ( $r = 0.22$ – $0.36$ ), but this signal largely overlaps with what Pythagorean already encodes. When Pythagorean is partialled out, most EPA partial correlations collapse to near zero: net EPA per play falls from  $r = 0.364$  to partial  $r = 0.057$  ( $p = 0.10$ ), and pass EPA drops from  $r = 0.301$  to partial  $r = 0.054$  ( $p = 0.12$ ).

**Two metrics show residual signal that is real but small.** Success rate differential retains a partial  $r = 0.118$  ( $p < 0.001$ ) and contributes the largest incremental  $R^2$  of any tested feature ( $\Delta R^2 = +0.012$  beyond Pythagorean baseline). Rush EPA differential shows partial  $r = 0.088$  ( $p = 0.011$ ,  $\Delta R^2 = +0.007$ ). These are statistically detectable relationships, but the practical gain — approximately 0.03–0.05 wins MAE improvement — is below the resolution of the discrete grid search (0.05 weight increments) and does not survive the train/validation cross-validation framework. Whether finer-grained optimization or longer training

histories would unlock this signal is a question for future work.

**Table 5c: EPA mechanism — partial correlations and incremental  $R^2$**

| Feature             | Raw r  | Partial r ( | Pythagorean) | p      |
|---------------------|--------|-------------|--------------|--------|
| Pythagorean edge    | +0.361 | +0.361      | <0.001       | —      |
| Point diff/game     | +0.363 | +0.046      | 0.19         | +0.002 |
| Net EPA/play        | +0.364 | +0.057      | 0.10         | +0.003 |
| Pass EPA diff       | +0.301 | +0.054      | 0.12         | +0.003 |
| Rush EPA diff       | +0.232 | +0.088      | 0.011        | +0.007 |
| Success rate diff   | +0.360 | +0.118      | <0.001       | +0.012 |
| Explosive play diff | +0.221 | +0.023      | 0.51         | <0.001 |
| Turnover margin     | +0.211 | −0.028      | 0.42         | +0.001 |

This is a null result in the grid-search sense, not a refutation of EPA as a metric. The evidence suggests EPA metrics largely recapture the season-level quality signal already encoded in Pythagorean. The residual information in success rate and rush EPA is real but too small to survive the discrete optimization and cross-validation framework used here.

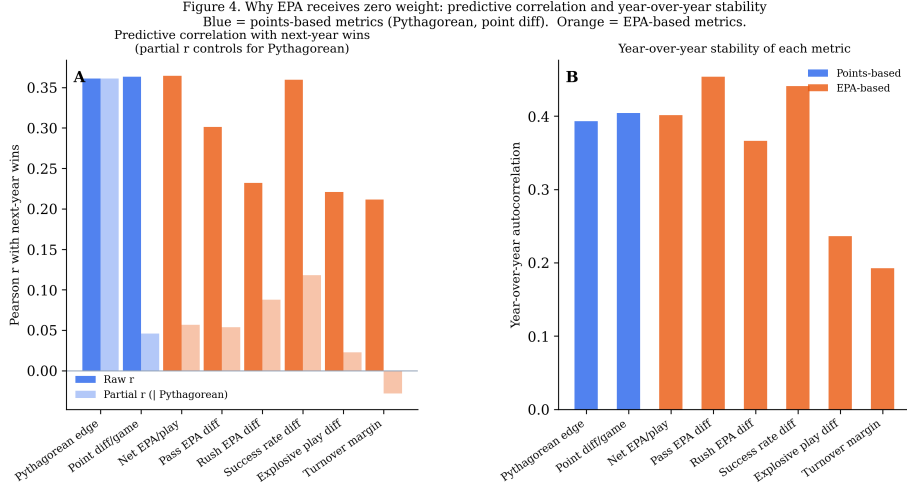


Figure 4: Figure 4. EPA mechanism analysis. Panel A: raw predictive correlation (solid bars) and partial correlation controlling for Pythagorean (lighter bars) for each metric. Panel B: year-over-year autocorrelation. Most EPA metrics are as stable or more stable than Pythagorean; the null result reflects multicollinearity, not noisiness.

## 5.8 Enhancement Tests — Principled Null Results

**Table 6: Investigation of potential predictive enhancements**

| Enhancement tested                    | Proposed mechanism                                 | Full MAE $\Delta$ | Val MAE $\Delta$ (2022–25) | Result      |
|---------------------------------------|----------------------------------------------------|-------------------|----------------------------|-------------|
| Strength of Schedule (weight 0.0–0.3) | Recursive quality adjustment for opponent strength | 0.000             | +0.002                     | <b>Null</b> |
| Regime Shift (R = 0.65 vs 0.75)       | Lower regression for modern “post-parity” NFL      | +0.002            | +0.001                     | <b>Null</b> |
| Garbage-Time Filter (WP [0.05, 0.95]) | Remove non-competitive plays for Pythagorean       | +0.025            | +0.025                     | <b>Null</b> |

All tests use the same train/validation split. The null result for three independent enhancements is itself a finding.

The SOS null result is a result, not just a limitation. We tested a recursive SOS adjustment that re-weights opponent quality into each team’s efficiency estimate. It produced zero improvement (full MAE  $\Delta = 0.000$ ) and a small increase in validation error (+0.002). The most likely mechanism: the NFL’s parity-scheduling system creates an endogenous feedback loop where strong teams face harder schedules and weak teams face softer ones, meaning an SOS correction pushes in the right direction but by approximately the same amount that schedule difficulty already depressed or inflated the observed statistics.

## 5.9 Calibration

We divide all team-season predictions into six equal-sized buckets by projected win total and examine whether predictions are systematically biased in any range.

**Table 7: Calibration by projected win bucket (full sample, 2000–2025)**

| Projected win range | Observations | Avg projected | Avg actual | Avg bias | MAE  |
|---------------------|--------------|---------------|------------|----------|------|
| 4.8 – 6.7           | 139          | 6.12          | 6.36       | −0.24    | 2.35 |
| 6.7 – 7.5           | 138          | 7.09          | 6.99       | +0.10    | 2.37 |
| 7.5 – 8.1           | 138          | 7.80          | 8.07       | −0.27    | 2.47 |
| 8.1 – 8.7           | 138          | 8.41          | 8.07       | +0.34    | 2.39 |
| 8.7 – 9.5           | 138          | 9.10          | 9.12       | −0.02    | 2.47 |
| 9.5 – 11.5          | 139          | 10.06         | 9.96       | +0.10    | 2.22 |

Biases are small (under 0.35 wins) in every bucket with no systematic directional pattern.

### 5.10 Directional Accuracy and Profitability

Across all WARPS bets with edge 0.5 wins (325 bets, 2003–2020), directional hit rate is 47.4% — below the 52.4% break-even at −110 juice. Undifferentiated betting on any WARPS signal is not profitable after vig. Filtering to 3-model consensus at 1.5 win edge raises the directional hit rate to **52.6%** (+9.5% ROI, 19 bets). The 19-bet sample is far too small for reliable inference; this should be treated as exploratory.

**Table 8: Profitability simulation against Vegas win totals (2003–2020)**

| Model                    | Min edge        | Bets      | Win%         | ROI          |
|--------------------------|-----------------|-----------|--------------|--------------|
| WARPS v1.8               | 0.5 wins        | 325       | 47.4%        | −9.6%        |
| WARPS v1.8               | 1.0 wins        | 155       | 46.7%        | −11.3%       |
| WARPS v1.8               | 1.5 wins        | 55        | 50.0%        | −5.4%        |
| <b>WARPS v1.8</b>        | <b>2.0 wins</b> | <b>8</b>  | <b>50.0%</b> | <b>+0.9%</b> |
| Pythagorean              | 2.0 wins        | 95        | 40.4%        | −20.4%       |
| <b>3-model consensus</b> | <b>1.5 wins</b> | <b>19</b> | <b>52.6%</b> | <b>+9.5%</b> |

The main finding: Vegas is largely efficient against publicly available efficiency metrics. WARPS’s better calibration is evident at extreme edges — at 2.0 win disagreements with Vegas, WARPS breaks even while Pythagorean loses at −20.4% ROI. Regression-to-mean correction prevents systematic overconfidence at the tails. Vegas wins.

### 5.11 2026 Season Consensus Screen

The three-model consensus screen identifies teams where all three WARPS versions agree on direction with individual edge 1.5 wins. Current-season applications and full projection tables are available at [anonymous repository — provided upon acceptance].

## 6. Discussion

### 6.1 Robustness, Stability, and the Limits of Complexity

The central finding of this investigation is not the specific parameter values selected. It is the pattern of what did and did not improve forecasts.

**Finding A: Simple points-based metrics dominate.** Pythagorean win expectation and raw point differential — both computed from aggregate scores — outperform a seven-metric composite that includes EPA-based efficiency measures. The more contextually precise metrics add nothing once the blunt instrument of points scored and allowed is included.

**Finding B: Coefficient tuning adds noise, not signal.** True walk-forward retraining selected varied configurations across all 16 independent windows (w\_pyth: 0.50–1.00, median=0.57; R: 0.55–0.75, median=0.68) — evidence that the training objective surface is too flat to identify a consistently superior configuration. Yet the fixed champion outperformed the window-specific optimum out-of-sample in 12 of 16 trials, by an average of 0.012 wins. The mechanism is straightforward: when configurations are separated by fractions of a thousandth of a win on training data, window-specific optimization fits noise. A fixed well-chosen configuration avoids this. All 210 tested configurations (100%) fall within the basin — every reasonable parameter choice produces essentially the same forecast.

**Finding C: The forecasting relationship is stable across time.** The fixed champion configuration outperformed window-specific optimization in 12 of 16 out-of-sample tests spanning 2010–2025, across very different training window sizes (10–25 seasons) and NFL eras. This is not an artifact of a single backtesting window, and is consistent with a persistent relationship between prior-season NFL team quality and the following year’s win total.

**Finding D: Complexity failed to improve forecasts.** Three independent model extensions — SOS adjustment, era-aware regression, garbage-time filtering — each produced null results. A null from any one test could be attributed to incorrect specification or insufficient statistical power. Null results from three independent interventions, each motivated by sound domain logic, suggest these mechanisms did not provide incremental predictive value within the framework tested.

The appropriate response to these findings is not to add more components. Repeated enhancement attempts did not materially improve forecasting accuracy.

## 6.2 The EPA Paradox

EPA has become the reference metric in contemporary NFL analytics for good reason: it is contextually fair, play-level efficient, and harder to game than raw yardage. It is precisely because EPA is sophisticated that its zero weight in the representative champion configuration is noteworthy.

The most compelling explanation is a temporal horizon mismatch. EPA’s contextual precision is valuable for evaluating individual plays and game decisions. At the season level, however, team quality is already encoded in aggregate scoring outcomes — and Pythagorean expectation processes that information nonlinearly in a way that empirically captures year-over-year persistence better than



EPA per play does.

A second explanation is noise accumulation. A team's EPA per play is influenced by game state, playcalling in obvious situations, and garbage-time volume. These introduce season-to-season variance that does not reflect underlying quality changes. Raw point differential and Pythagorean expectation, being aggregates of final scores, are less sensitive to in-game strategy patterns.

Whatever the mechanism, the finding is clear: EPA did not improve preseason win-total forecasts in this framework. This does not mean EPA is wrong. It means it is not incrementally useful here, given what points-based metrics already encode.

### 6.3 Why Pythagorean Dominates

The finding that Pythagorean win expectation is the most valuable signal is consistent with the broader sports analytics literature. Actual win-loss records contain substantial luck: close games, fumble bounces, and tipped passes create variance around the true quality of a team. Pythagorean expectation averages out this game-level variance by focusing on cumulative points scored and allowed, which are harder to sustain artificially over a full season.

An apparent tension in the results warrants clarification. WARPS beats the Pythagorean baseline in 25 of 26 seasons, yet the full-sample minimum occurs at  $w\_pyth=1.00$  with no point-differential supplement. These findings are not contradictory because they compare different benchmarks.

The season-by-season comparison uses raw Pythagorean expectation as the baseline — that is, Pythagorean win expectation with no regression toward the league mean ( $R=1.00$ ). The full-sample minimum, by contrast, applies regression toward the league mean ( $R=0.75$ ) before generating win projections. The evidence suggests that regression is responsible for most of the predictive improvement over raw Pythagorean. Once regression is applied, the objective surface becomes extremely flat: pure Pythagorean, the representative 75/25 blend, and many nearby configurations produce nearly identical forecast accuracy.

This interpretation reinforces the central conclusion of the paper. The primary contribution is not identifying a precise optimal blend between Pythagorean expectation and point differential. Rather, it is demonstrating that a regressed points-based signal is stable, robust, and resistant to meaningful improvement from the enhancements tested here.

### 6.4 Why Simplicity Wins

Points scored and allowed are among the most information-dense statistics in team sports. A final score aggregates hundreds of individual plays — offensive execution, defensive resistance, special teams, and in-game strategy — into a single number. Pythagorean expectation then non-linearly transforms that aggregate in a way that downweights fluky close-game outcomes and upweights

sustained dominance. The result is a single statistic that already encodes much of what any advanced metric is trying to measure.

Most modern NFL efficiency metrics — EPA per play, success rate, explosive play rate — are computed from the same play-by-play events that ultimately determine the final score. They are, in a precise sense, upstream components of points. Once a season’s aggregate scoring is known, these play-level summaries provide limited additional information about a team’s underlying quality, because that quality has already been expressed in the score. The partial correlation analysis in §5.7 confirms this empirically: EPA metrics lose almost all predictive power once Pythagorean expectation is controlled for.

This explains why the objective surface is flat across a wide parameter range, and why repeated enhancement attempts failed. The forecasting problem is not under-specified — it has a dominant signal, and the challenge is identifying it. Once Pythagorean expectation captures that signal, alternatives that appear richer in theory offer diminishing returns in practice. The finding is not that complexity is wrong. It is that the information complexity attempts to add has already been processed and discarded in the aggregation to final scores.

### **6.5 Why Point Differential Adds Value (Marginally)**

When we have 22 training seasons, the optimizer identifies a role for raw point differential alongside Pythagorean. The two metrics carry overlapping but not identical information. Pythagorean applies a 2.37 exponent that non-linearly up-weights blowout margins. A team that wins every game by 3 points has the same total point differential as one that alternates blowout wins with close losses, but their Pythagorean scores differ substantially. Raw point differential, being linear, treats these teams more similarly. One representative configuration suggests a blend performs slightly better: Pythagorean’s non-linear sensitivity is valuable, but pure Pythagorean can overweight seasons where blowout margins may not persist.

### **6.6 The 2014 Exception**

The only season where WARPS underperformed Pythagorean was 2014 (WARPS MAE 2.094 vs Pythagorean 2.018, margin 0.076 wins). The 2014 exception illustrates that efficiency-based models remain vulnerable to personnel and organizational changes not represented in prior-season statistics.

### **6.7 NFL Regime Volatility — 2024 and 2025**

The 2024 and 2025 seasons produced the highest WARPS MAEs in the 26-season sample (3.01 and 2.67 respectively). The fat-tail errors are concentrated in two identifiable groups: (1) sustained-excellence teams that exceeded what any regression-toward-mean model can capture; and (2) rapid collapse teams whose declines were associated with major personnel and organizational changes not represented in the model. Critically, the Vegas market also produced its

second-worst MAE in 2024 (2.86 wins), confirming that 2024–2025 represented an industry-wide forecasting challenge. Diagnostic analysis suggests these large residuals were primarily associated with factors such as coaching turnover and quarterback disruption. These post-hoc observations are descriptive rather than causal and should be interpreted as hypotheses for the observed fat-tail errors. Conditional regression schemes for sustained-excellence franchises represent a promising direction for future work.

## 6.8 The Role of the Consensus Screen

The three-model consensus screen is designed to reduce false positives. Each model version was trained with different data windows or search procedures, making their errors partially independent. When all three agree, the signal is more robust than any single model alone.

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## 7. Limitations

**Personnel changes are not modeled.** The model uses only prior-season on-field statistics. It does not incorporate quarterback changes, major free agency moves, or coaching staff turnover. A team losing its franchise quarterback can shift true talent by several wins in ways no efficiency metric can capture.

**Small validation window.** The held-out validation period is only four seasons (2022–2025). While statistically significant, conclusions would be strengthened by additional future seasons.

**Historical era effects.** The NFL changed significantly over the 26-season study period. A model trained on 2000–2021 data implicitly assumes structural shifts average out over the training window.

**Market efficiency.** Vegas preseason win totals already incorporate public information, including team efficiency statistics. The edges we identify may be smaller in practice after vig, and this paper does not claim to identify profitable betting opportunities — only statistically significant forecast improvements over statistical baselines.

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## 8. Conclusion

This investigation began as a search for alternatives to Pythagorean expectation. The evidence consistently pointed elsewhere: Pythagorean expectation is not a baseline to be replaced but the dominant forecasting signal, and the primary contribution of this work is evidence for the stability and robustness of that fact.

True walk-forward retraining across 16 independent windows (2010–2025) selected varied configurations (median  $w_{\text{pyth}}=0.57$ , range 0.50–1.00), reflecting a training objective surface so flat that noise determines the nominal optimum each window. Despite this, the fixed champion outperformed window-specific optimization out-of-sample in 12 of 16 trials, by an average of 0.012 wins. A 2D parameter heatmap confirms this flatness: 100% of tested configurations fall within 0.05 wins of the champion. WARPS beats the Pythagorean baseline in 25 of 26 seasons (96%), with statistically significant improvements at  $p < 0.0001$  on the full 26-season backtest and  $p < 0.01$  on the held-out validation window.

Equally important is what failed to help: EPA metrics, schedule strength adjustment, era-aware regression, and garbage-time filtering each produced null results once points-based signals were included. Complexity did not improve forecasts. The model that survived repeated enhancement attempts is the finding.

The betting market remains largely efficient: neither WARPS nor Pythagorean generates positive ROI at standard thresholds, confirming that publicly available statistical signals are already priced in. WARPS’s better calibration at extreme edges (breaking even at 2.0 win disagreements where Pythagorean loses at −20.4% ROI) confirms the value of regression-to-mean correction, but does not constitute a market edge. Vegas wins.

Future work should focus on extending the walk-forward window to additional out-of-sample seasons as they accumulate, and investigating whether the EPA null result holds across shorter forecast horizons (e.g., eight-week mid-season projections). One promising direction is conditional regression schemes that allow sustained elite franchises to revert more slowly toward the mean. Preliminary testing suggested modest improvements, but this extension was excluded from the primary framework to preserve model parsimony.

These findings should be interpreted as evidence of persistent forecasting relationships across the observed historical sample rather than proof of a universal or invariant law of NFL outcomes.

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## Appendix A — Raw Walk-Forward Results (2010–2025)

Full annual records for reproduction (true retraining from raw play-by-play data).  $\Delta = \text{OOS MAE}(\text{champion}) - \text{OOS MAE}(\text{optimal})$ . **Negative delta = champion wins** (lower champion MAE).  $w_{\text{pd}} = 1 - w_{\text{pyth}}$  for each row.

| Year | N   | $w_{\text{pyth}}$ | R    | Train<br>MAE | OOS<br>Champ | OOS<br>Opt | Delta   |
|------|-----|-------------------|------|--------------|--------------|------------|---------|
| 2010 | 318 | 0.50              | 0.55 | 2.4087       | 2.4024       | 2.3860     | +0.0161 |
| 2011 | 350 | 0.50              | 0.55 | 2.4054       | 2.1068       | 2.1456     | −0.0388 |
| 2012 | 382 | 0.65              | 0.60 | 2.3811       | 2.5347       | 2.5323     | +0.0024 |
| 2013 | 414 | 1.00              | 0.60 | 2.3933       | 2.3640       | 2.3675     | −0.0035 |

| Year           | N   | w_pyth      | R           | Train<br>MAE | OOS<br>Champ | OOS<br>Opt   | Delta         |
|----------------|-----|-------------|-------------|--------------|--------------|--------------|---------------|
| 2014           | 446 | 0.95        | 0.60        | 2.3908       | 2.0939       | 2.1867       | −0.0928       |
| 2015           | 478 | 0.70        | 0.65        | 2.3735       | 2.3005       | 2.3097       | −0.0092       |
| 2016           | 510 | 0.70        | 0.65        | 2.3719       | 2.4253       | 2.4316       | −0.0063       |
| 2017           | 542 | 0.70        | 0.65        | 2.3749       | 2.2170       | 2.2387       | −0.0217       |
| 2018           | 574 | 0.60        | 0.70        | 2.3653       | 2.0910       | 2.1044       | −0.0134       |
| 2019           | 606 | 0.50        | 0.70        | 2.3508       | 2.2118       | 2.2131       | −0.0013       |
| 2020           | 638 | 0.50        | 0.70        | 2.3445       | 2.7799       | 2.7879       | −0.0080       |
| 2021           | 670 | 0.55        | 0.70        | 2.3655       | 1.9378       | 1.9474       | −0.0096       |
| 2022           | 702 | 0.50        | 0.70        | 2.3457       | 2.4599       | 2.4466       | +0.0133       |
| 2023           | 734 | 0.50        | 0.70        | 2.3494       | 1.8981       | 1.8987       | −0.0006       |
| 2024           | 766 | 0.50        | 0.70        | 2.3301       | 3.0125       | 3.0360       | −0.0235       |
| 2025           | 798 | 1.00        | 0.75        | 2.3587       | 2.6732       | 2.6640       | +0.0092       |
| <b>Summary</b> |     | <b>0.57</b> | <b>0.68</b> |              | <b>2.376</b> | <b>2.388</b> | <b>−0.012</b> |
|                |     | <b>med</b>  | <b>med</b>  |              | <b>avg</b>   | <b>avg</b>   | <b>avg</b>    |

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*Data and code: [anonymous repository — provided upon acceptance] Model version: WARPS-NFL v1.8 · Training window: 2000–2021 · Validation window: 2022–2025 Paper version: v2.2 · Last updated: June 2026*